Josiah Lucas

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SUMMARY

Computer Science graduate focused on backend engineering and applied AI. Built and deployed AlphaTracker, a full-stack trading journal (Java/Spring Boot, PostgreSQL, React/TypeScript, AWS S3) with 65+ automated tests, and shipped a Python AI assistant on top of it, validated with a 50-question eval harness. Merged contributor to Alibaba's open-code-review (30k+ stars).

EDUCATION

San Francisco State University — B.S. Computer Science *August 2026*

- Relevant Coursework: Deep Learning, Systems Programming, Data Structures & Algorithms
- Fellowships: CodePath Technical Interview Prep (TIP) — Completed

TECHNICAL SKILLS

Languages: Java, TypeScript, Go, Python, C++

Backend & Web: Spring Boot, FastAPI, Spring Data JPA, REST APIs, JWT authentication, React, Node.js

Cloud & DevOps: AWS (S3, IAM), Docker, GitHub Actions, Railway, Vercel

Databases: PostgreSQL (Supabase)

Testing: JUnit 5, Mockito, Spring Boot test slices (@DataJpaTest, @WebMvcTest)

AI/LLM: Gemini API, Pydantic structured-output validation, multimodal (image + text) pipelines

PROJECTS

AlphaTracker (Live Demo) (GitHub) | Trading Journal and AI Assistant for Prop-Firm Traders *May 2026 - October 2026*

Stack: Java (Spring Boot), PostgreSQL, TypeScript, React, AWS S3, Docker, Python, FastAPI, Gemini API

Core Platform

- Engineered a Spring Boot REST API that derives all P&L, commission, and drawdown server-side from a single source of truth for 12 CME futures contracts, rejecting unknown tickers to eliminate client-side calculation errors.
- Designed a **trailing drawdown engine** that replays each account's closed-trade history to compute its high-water mark and live floor, supporting end-of-day and per-trade trailing modes across multiple prop accounts.
- Migrated production file storage from local disk to **AWS S3** through a pluggable storage interface with zero changes to calling code, and secured every endpoint with JWT authentication and per-request ownership checks.
- Caught a date-range bug that double-counted trades and a password-hash leak in API responses by writing **65+ JUnit 5/Mockito tests** against real PostgreSQL across repository, service, and controller layers.
- Deployed the full stack to production with Docker on Railway, Supabase Postgres, and Vercel, diagnosing IAM, CORS, and environment configuration issues from live server logs.

AI Assistant

- Shipped an **AI trade-review assistant**: a Python FastAPI agent using Gemini function calling to answer natural-language questions through read-only, user-scoped tools on the Spring API, deployed as its own Railway service behind a React chat panel.
- Implemented a grounding guard that verifies every figure in a model response against tool outputs and retries ungrounded answers, reducing the correction rate from **12% to 2.7%**.
- Validated the agent with a deterministic eval harness of 50 questions across 9 categories, including prompt injection, scored against an independent Python oracle: 148/150 passed, **18/18 on a blind held-out set**, and 100% tool-selection accuracy.
- Hardened the service for public traffic with per-user and global sliding-window rate limits returning 429 with Retry-After, verified by a concurrency test that fails when the lock is removed.

Multi-Modal Evidence Review AI Agent (GitHub) | HackerRank Orchestrate Hackathon *May 2026 - June 2026*

Python, Gemini 2.5 Flash, Pydantic, Git

- Built an asynchronous Python pipeline on **Gemini 2.5 Flash** that analyzes image and text evidence for insurance fraud review.
- Validated every model response against Pydantic schemas, converting LLM output into type-safe JSON and rejecting malformed responses.
- Added guardrail checks in the execution path to catch hallucinated and out-of-scope outputs before they reached the final decision.

Open Source Contribution (GitHub) *September 2026*

Alibaba open-code-review (30k+ ★) | *Go, HTML/CSS, Open Source* **| Merged Pull Request**

- Restyled the coverage and token-usage report in a Go-templated tool used across the project's review pipeline, preserving its data bindings across light and dark themes and merging after code review.

EXPERIENCE

Corica Park — *Guest Experience Host* *May 2025 – Present*

- Booked tee times, kept groups on schedule, and helped run tournaments as a first point of contact for guests, working 35+ hours a week while completing a full-time Computer Science degree.